NAND TO TETRIS ASSIGNMENT

Assessment Number: 1

Course: B.Tech CSE AI and ML

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Section: F

Subject: NAND to Tetris

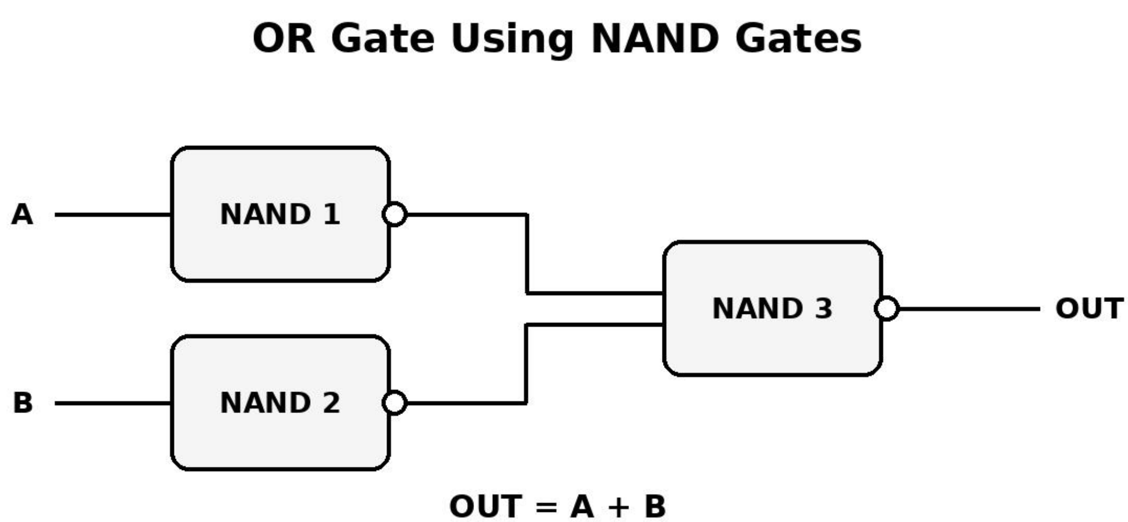
Question 1 :

Implement NOT, AND, and OR gates using only NAND gates in HDL. Draw the logic diagram for each and write the Boolean expression. Run the provided test scripts and capture the simulation output.

Answer:

1. NOT Gate

➢ Logic Diagram

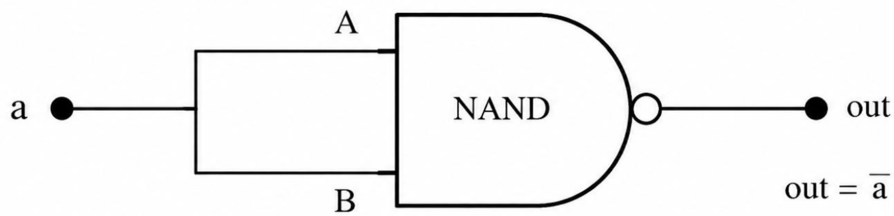


Boolean Expression: NOT(a) = a NAND a = a'

➢ Truth Table of NOT Gate

NOT Gate Using NAND Gate

Logic Diagram:

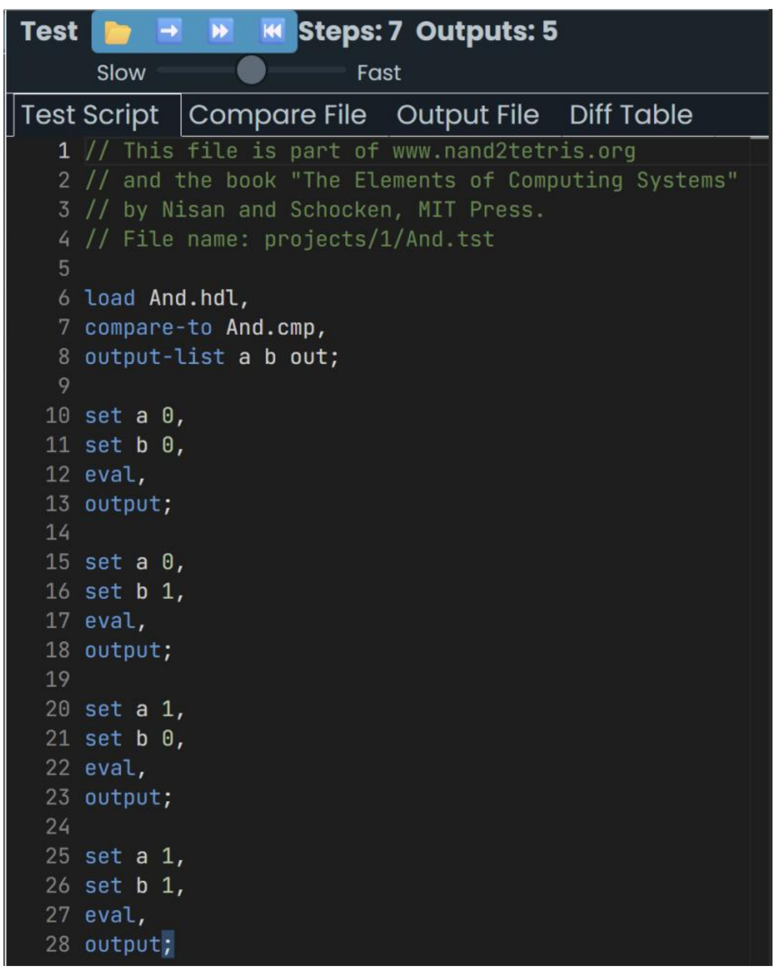


Both inputs of the NAND gate are connected to the same input ‘a’. Therefore, the output is the complement of a.

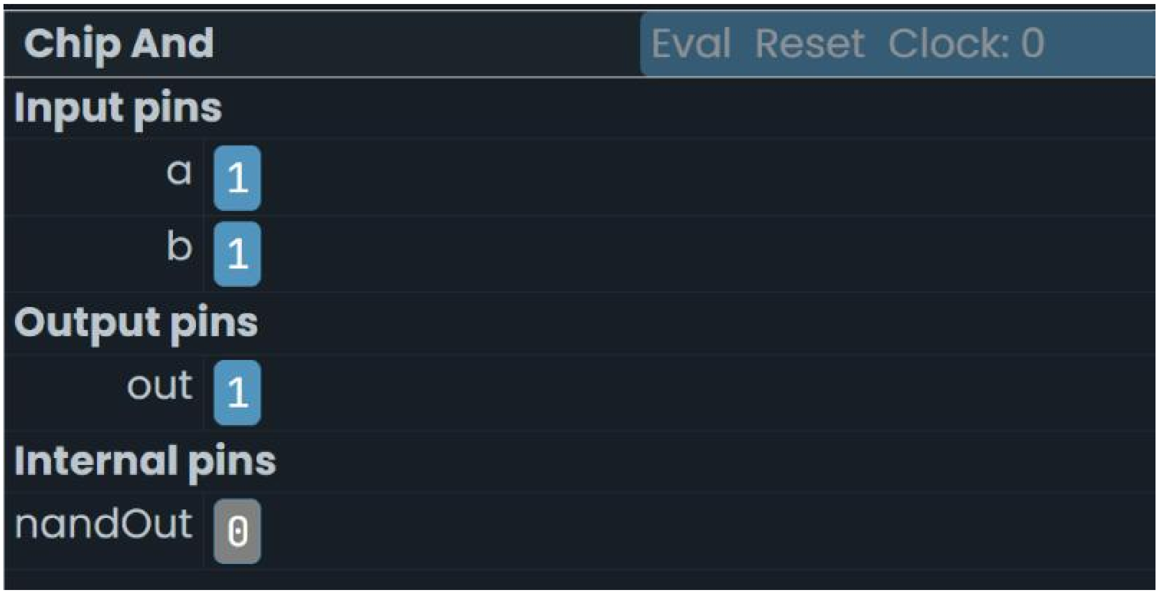
Boolean Expression:

$$\boxed{\text{NOT(a)} = \text{a NAND a} = \overline{\text{a}}}$$

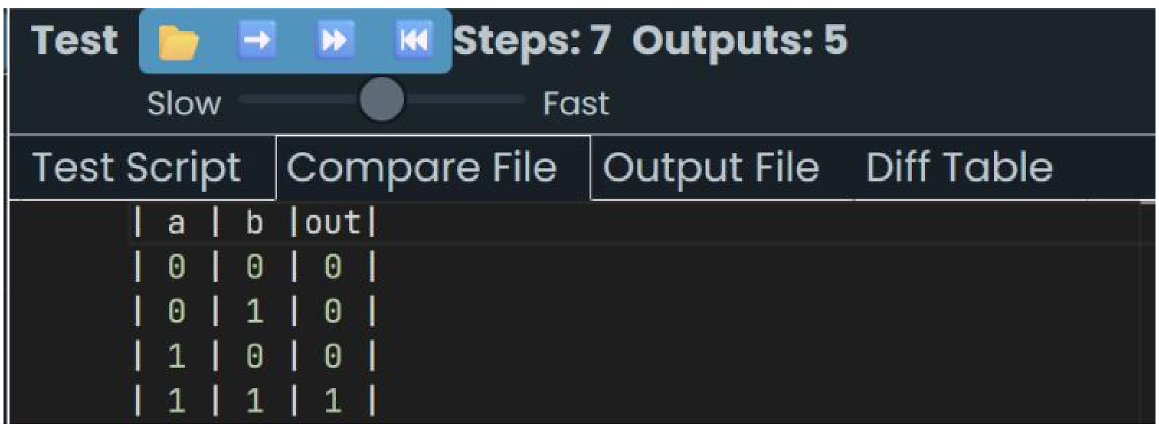
➢ HDL Dashboard



➢ **Test Script**

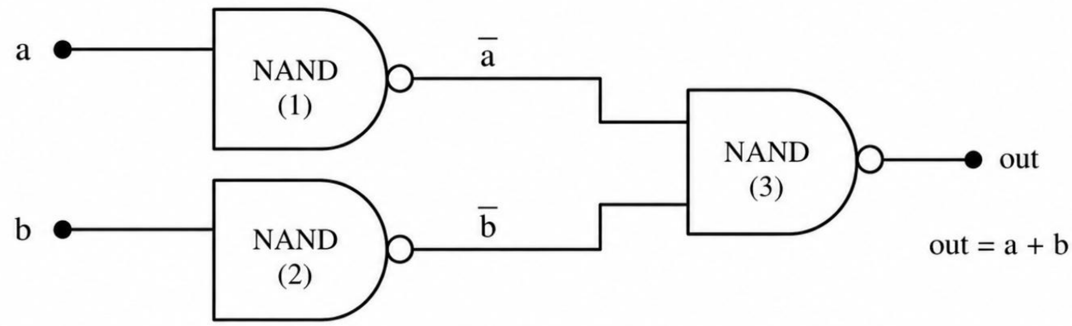


➢ **Pins**



➢ **Comparison Table**

Logic Diagram:



The first NAND gate inverts ‘ a ’ to $\bar{a}$.

The second NAND gate inverts ‘ b ’ to $\bar{b}$.

The third NAND gate computes the NAND of $\bar{a}$ and $\bar{b}$.

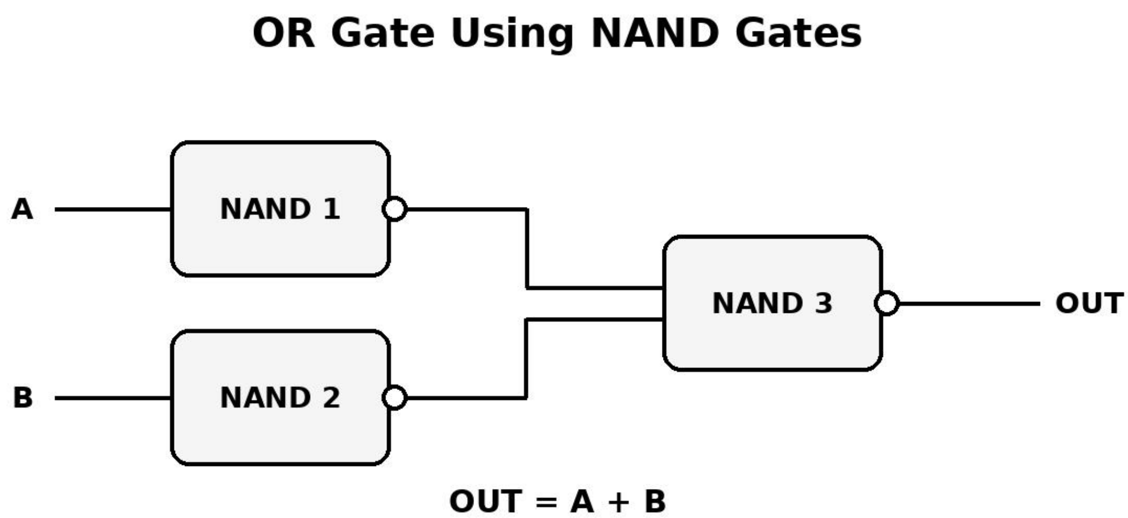
By De Morgan’s law, $\overline{a \cdot b} = a + b$.

Boolean Expression:

$$\boxed{\mathrm{OR}(a, b) = \overline{a \cdot b} = a + b}$$

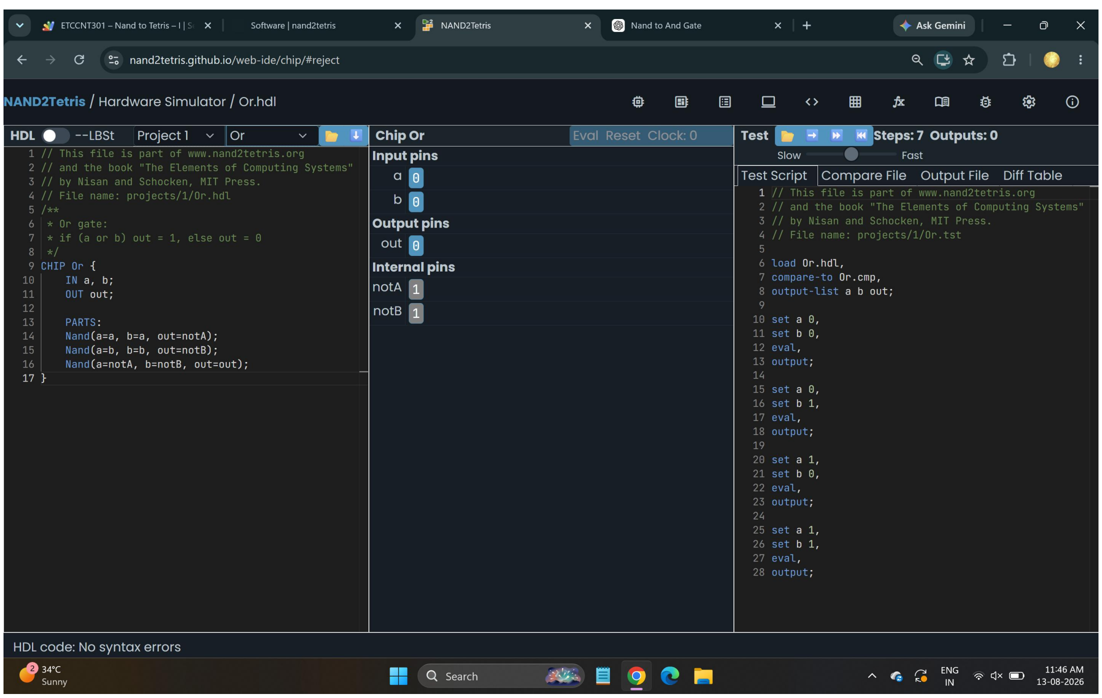
2. AND Gate

➢ Logic Diagram

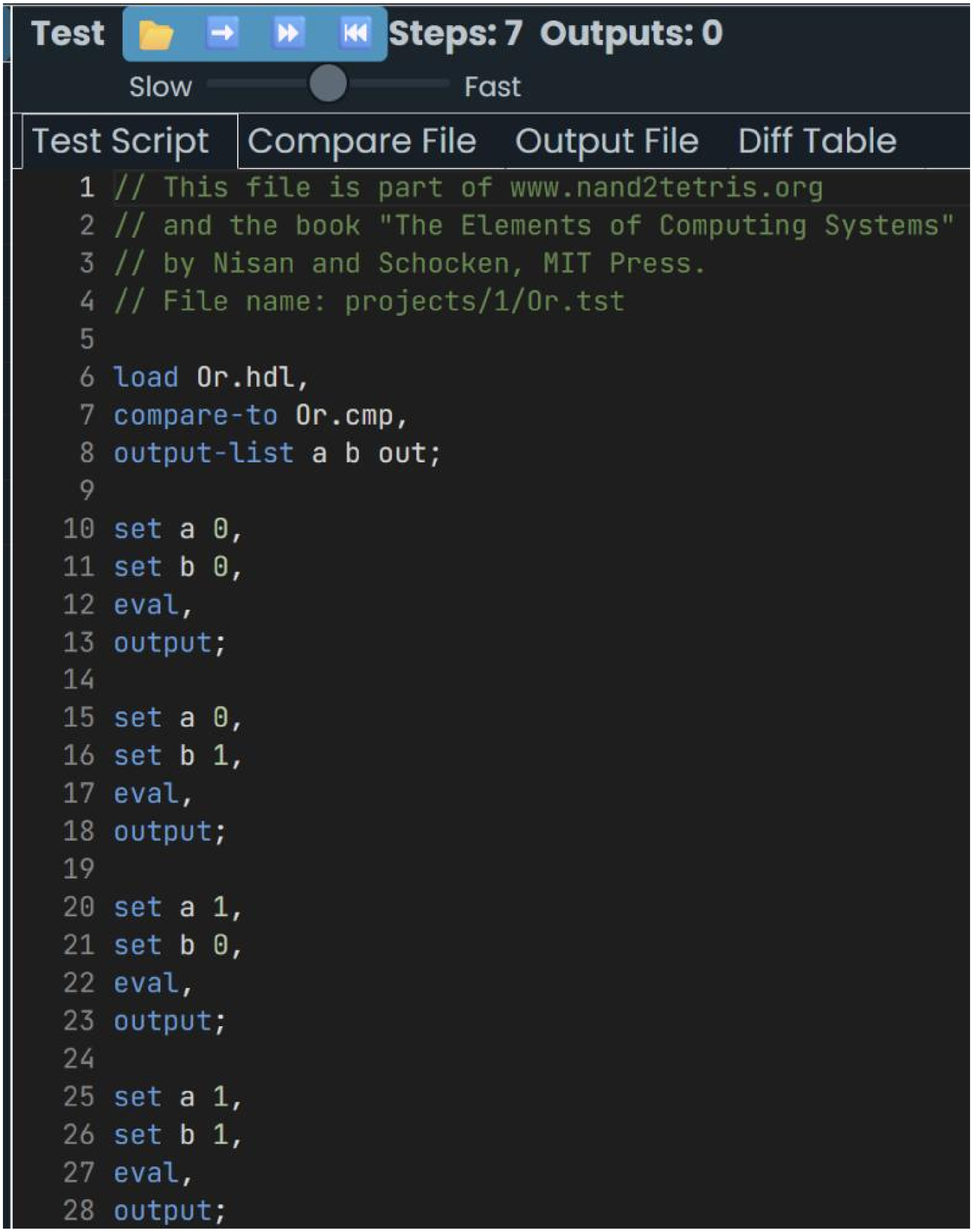


Boolean Expression: AND(a,b) = a · b

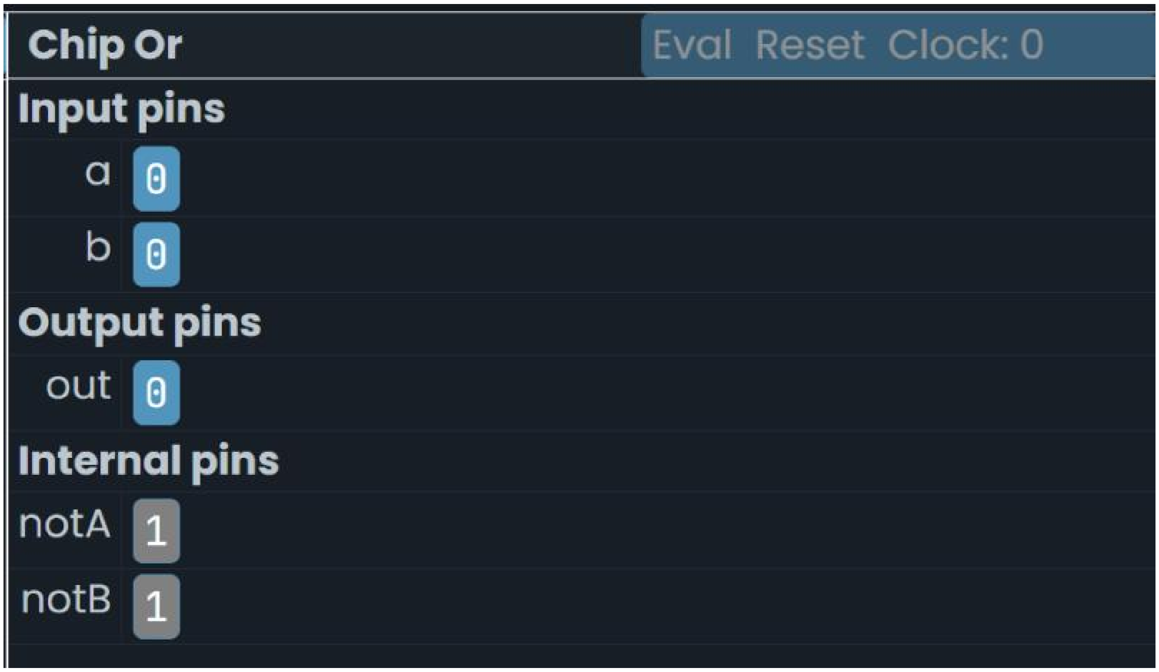
➢ Truth Table



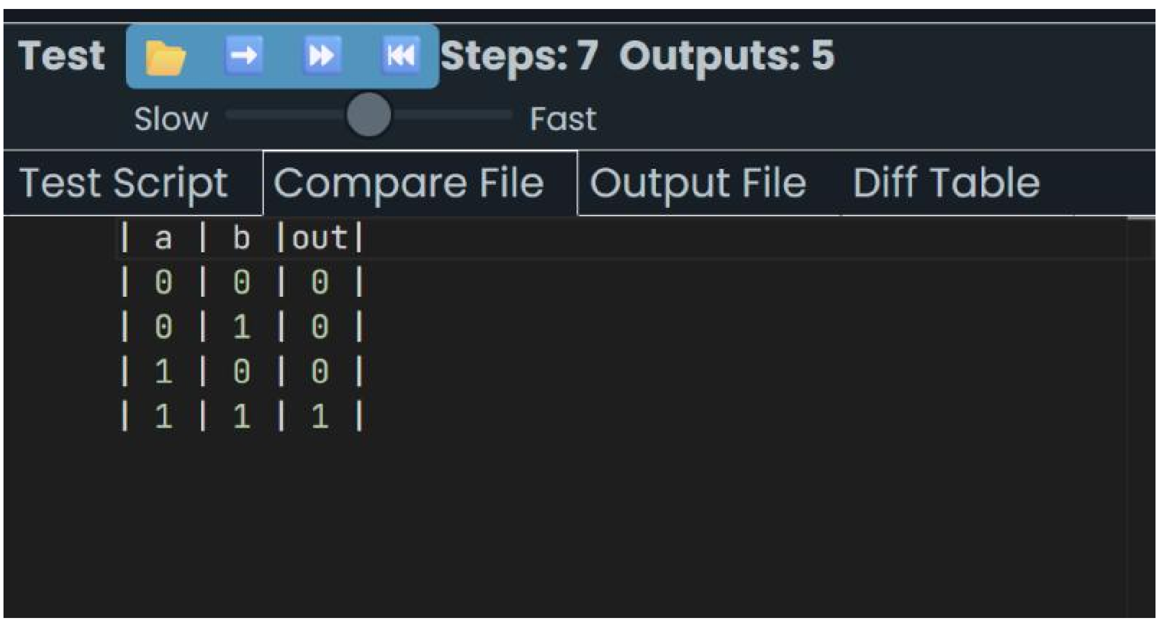
➢ HDL Dashboard


Test
Steps: 7 Outputs: 0
Slow
Fast
Test Script
Compare File
Output File
Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Or.tst
load Or.hdl,
compare-to Or.cmp,
output-list a b out;
set a 0,
set b 0,
eval,
output;
set a 0,
set b 1,
eval,
output;
set a 1,
set b 0,
eval,
output;
set a 1,
set b 1,
eval,
output;

➢ **Test Script**



➢ **Pins**



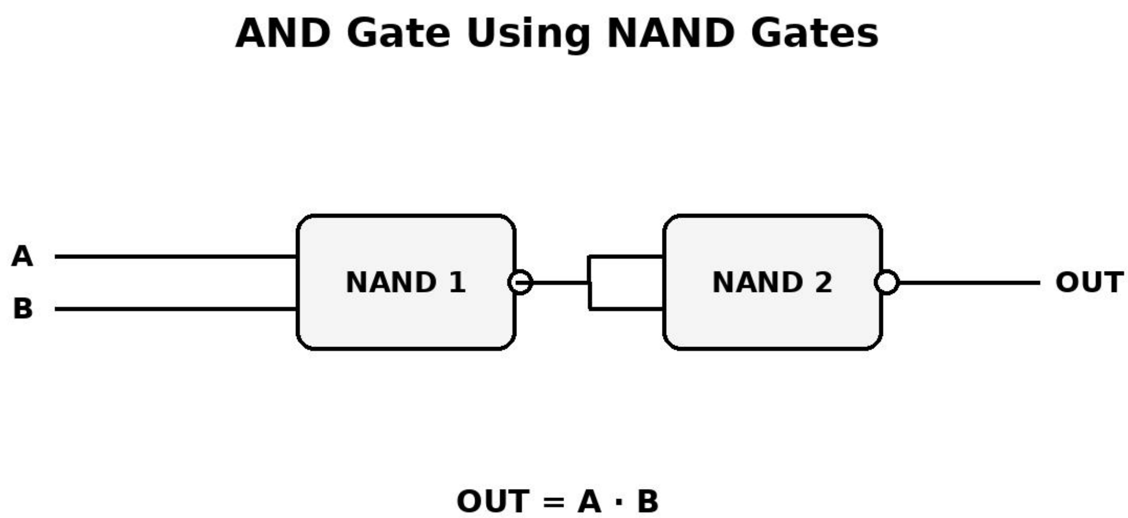
➢ **Comparison Table**

Truth Table

A (Input)	B (Input)	Y = $(\overline{AB}) = \overline{A}$
0	0	1
1	1	0

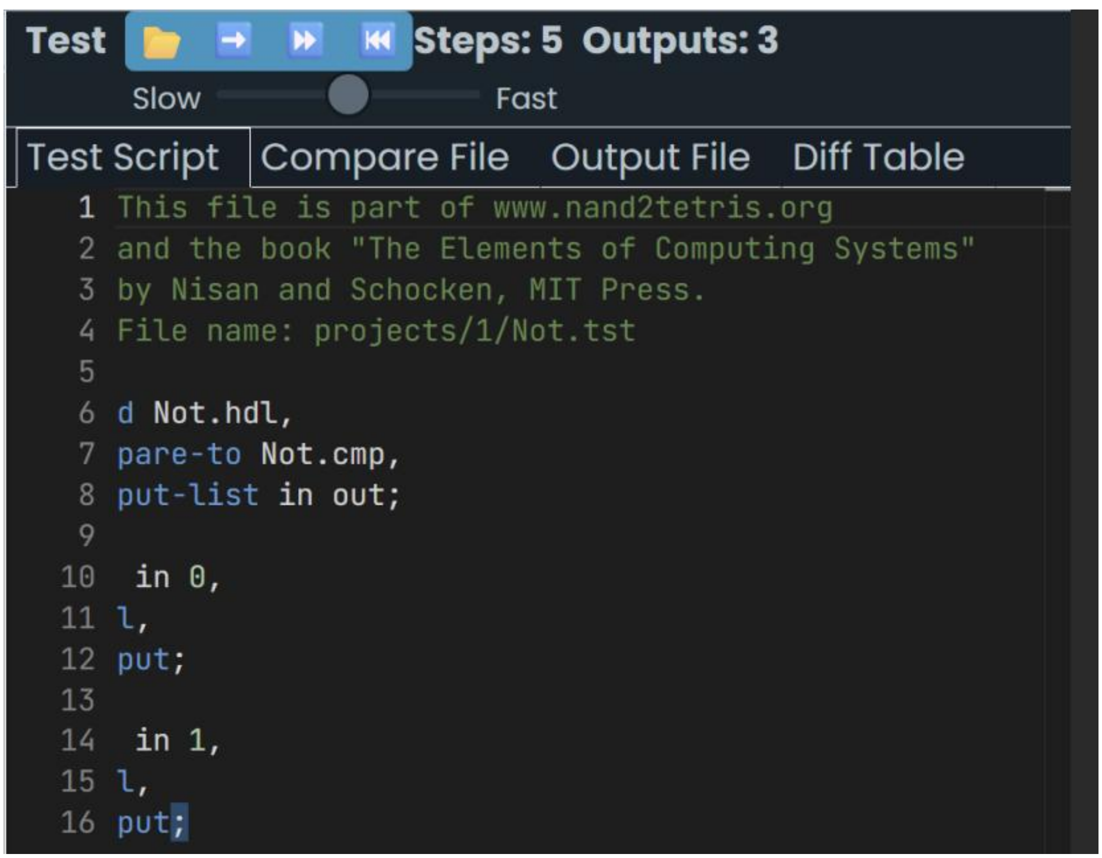
3. OR Gate

➢ Logic Diagram

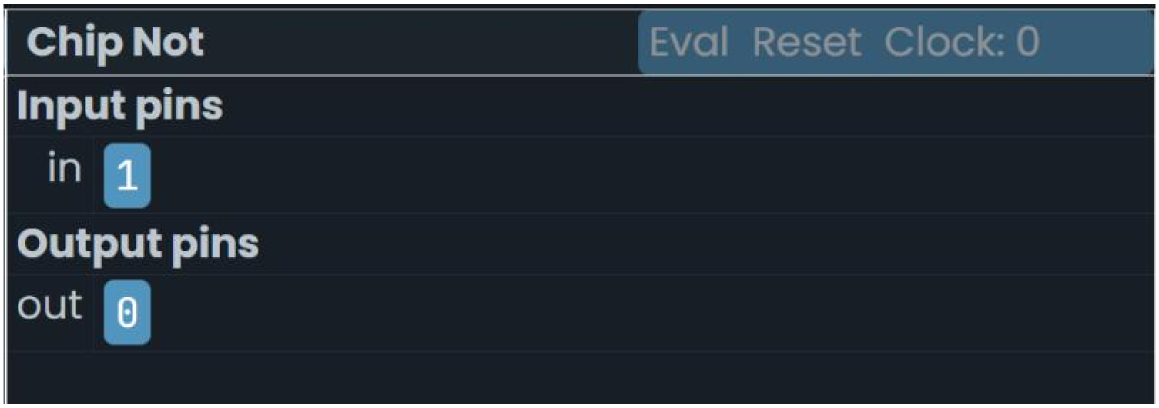


Boolean Expression: OR(a,b) = a + b

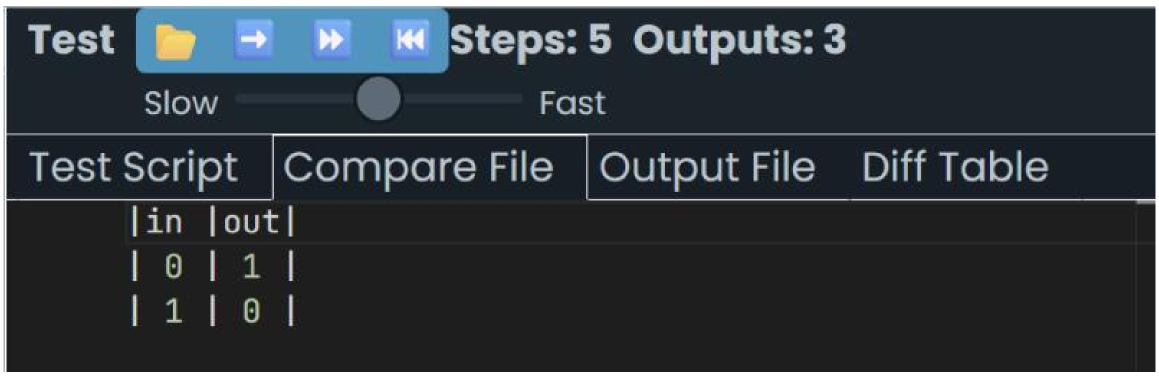
➢ Truth Table



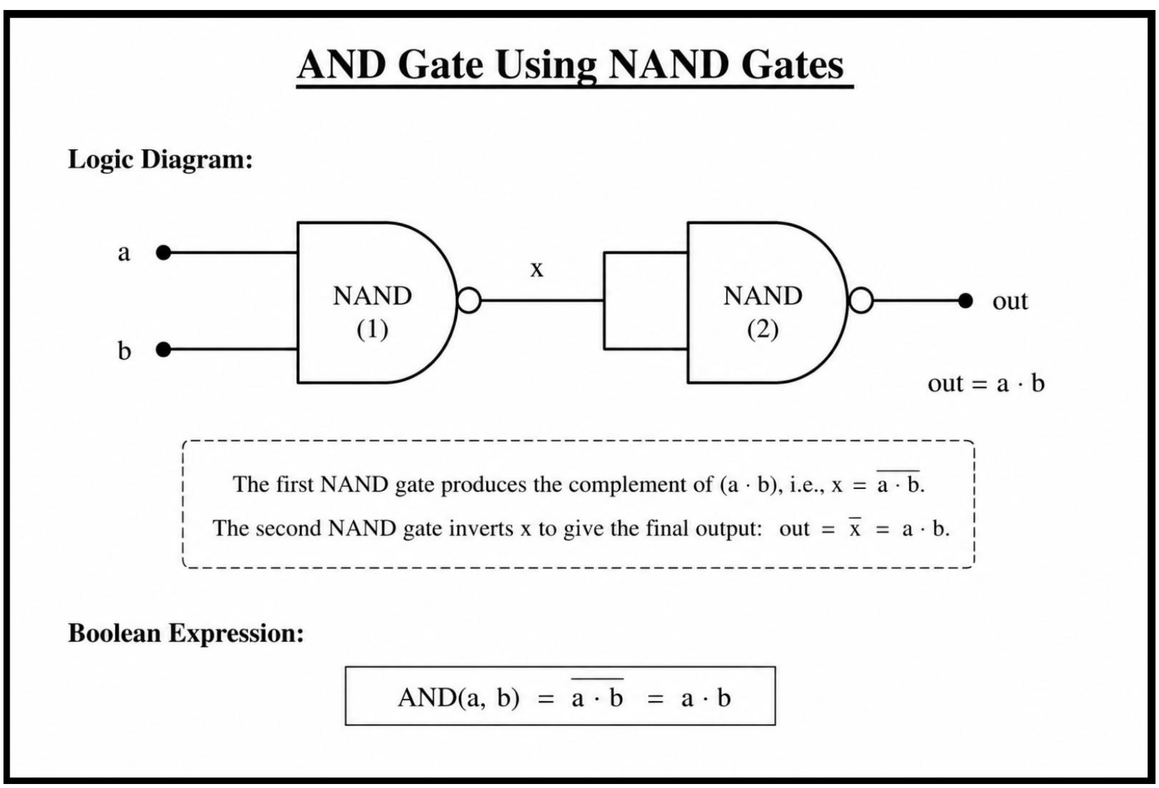
➢ HDL Dashboard



➢ **Test Script**



➢ **Pins**



➢ **Comparison Table**

Input A	Input B	Output $(A \cdot B)$
0	0	0
0	1	0
1	0	0
1	1	1

Conclusion

The NAND gate can be used as a universal logic gate to construct NOT, AND, and OR operations.